Atomic Structure

Answer sheet · 24 marks total

Section A — Sub-atomic particles

Proton relative charge = +1 (1)   Neutron relative mass = 1 (1)   Electron relative mass = very small / negligible (1)

Protons and neutrons are found in the nucleus (1)

[4 marks total]

Section B — Counting particles

B1. Neutrons = 27 – 13 = 14 (1)   Electrons = 13 (1)

B2. Neutrons = 40 – 20 = 20 (1)   Electrons = 20 (1)

Section C — Isotopes

C1. Atoms of the same element / same number of protons (1), with a different number of neutrons / different mass number (1).

C2. Both have 6 protons, so both are carbon (same atomic number = same element) (1). They have different numbers of neutrons, so different mass numbers, giving them different masses (1).

Section D — Electronic structure

D1. Nitrogen (7 electrons): **2,5** (2)

D2. Silicon (14 electrons): **2,8,4** (2)

D3. Potassium (19 electrons): fill 2, then 8, then 8 (= 18), 1 electron left starts a new shell (1)   **2,8,8,1** (1)

Section E — Development of the atomic model

E1. A ball/sphere of positive charge (1) with negative electrons embedded in it, like fruit in a pudding (1).

E2. Most alpha particles passed straight through the gold foil, showing that atoms are mostly empty space (1). A small number were deflected, and a very small number bounced back almost the way they came (1). This showed that most of the mass and all of the positive charge is concentrated in a tiny central region (1), leading to the nuclear model of the atom — a small, dense, positively charged nucleus with electrons around it (1).

Marking note: award method marks for correct working (e.g. neutrons = mass number – atomic number) even if the final answer contains an arithmetic error.